Lesson 3 Teacher's Key

Scratch

INSTRUCTOR ONLY - do not distribute to students

Level 1 - Complete Worked Solution

Two separate scripts on two separate sprites - this separation matters, since combining create-clone and delete-this-clone in the same "when I start as a clone" stack can freeze Scratch:

```
-- On the Player sprite:
when [space] key pressed
  create clone of [Bullet]

-- On the Bullet sprite:
when green flag clicked
  hide

when I start as a clone
  show
  go to x: (Player's x) y: (Player's y)
  forever
    change y by 10
    if <y position > 170> then
      delete this clone
```

Level 2 - Complete Worked Solution

Add a cooldown variable (can_shoot) on the Player so holding spacebar can't fire every single frame:

```
-- On the Player sprite:
when green flag clicked
  set [can_shoot] to true

when [space] key pressed
  if <can_shoot> then
    create clone of [Bullet]
    set [can_shoot] to false
    wait 0.2 seconds
    set [can_shoot] to true
```

Level 3 - All Worked Solutions

Every stretch-goal idea from the handout, fully worked - not just one. Useful for validating each idea actually works before class, and as a ready reference no matter which one a student ends up picking.

Sound: play on fire

Upload sfx_shoot.wav from this lesson's Assets folder, then add "start sound" right next to the clone-creation block - "start sound" doesn't block or loop, so it's fire-and-forget, the same property that made Lesson 2's beep-beep idea work:

```
when [space] key pressed
  if <can_shoot> then
    create clone of [Bullet]
    start sound [sfx_shoot]
    set [can_shoot] to false
    wait 0.2 seconds
    set [can_shoot] to true
```

Visual: give the bullet its own look

A quick "change size by" pulse on spawn reads clearly as a muzzle-flash-style pop, without needing a second costume:

```
when I start as a clone
  show
  set size to 150%
  go to x: (Player's x) y: (Player's y)
  wait 0.05 seconds
  set size to 100%
```

Structure: keep bullet behavior on the Bullet sprite

The Player's fire script should only ever trigger a shot (create clone, cooldown, sound) - everything about what a bullet actually does once it exists (appearing, moving, cleaning itself up) belongs on the Bullet sprite's own "when I start as a clone" script. Keeping this boundary clean is what let the Sound and Visual ideas above drop in with one line each, instead of needing to touch the Player's code.

Complete Level 3 solution (all 3 goals combined)

Everything above combined, across the Player and Bullet sprites:

```
-- Player sprite:
when green flag clicked
  set [can_shoot] to true

when [space] key pressed
  if <can_shoot> then
    create clone of [Bullet]
    start sound [sfx_shoot]
    set [can_shoot] to false
    wait 0.2 seconds
    set [can_shoot] to true

-- Bullet sprite:
when green flag clicked
  hide

when I start as a clone
  show
  set size to 150%
  go to x: (Player's x) y: (Player's y)
  wait 0.05 seconds
```

```
  set size to 100%
  forever
    change y by 10
    if <y position > 170> then
      delete this clone
```

Super-Stretch (not required): poll input instead of reacting to it, so firing can't interrupt movement

This is the worked fix for the known Scratch quirk noted in the Handout's Trouble section and this lesson's Sidebars: "when key pressed" scripts repeat by riding the operating system's own keyboard auto-repeat, and most operating systems only actively auto-repeat one key at a time - so pressing Space to fire while holding a direction key can interrupt that direction key's repeat stream, pausing movement until it's pressed again.

The fix is to stop depending on repeated key-press events entirely and poll real key state instead, using the Sensing category's "key ( ) pressed?" reporter inside a "forever" loop - the same technique Lesson 2's blink stretch goal already uses, now applied to movement and firing too. Because this reads whether a key is currently held down, once per frame, it's completely unaffected by the operating system's one-key-at-a-time auto-repeat limit.

Two separate "when green flag clicked" scripts on the Player, not one - movement polls independently from firing, so the "wait 0.2 seconds" fire cooldown pauses only the fire script's own thread, not movement. Combining both into a single forever loop would make movement visibly freeze for the whole cooldown every time you fire, trading one bug for another:

```
-- Player sprite, movement script (replaces the 4 "when [arrow] key
pressed" scripts):
when green flag clicked
  forever
    if <key [right arrow] pressed?> then
      change x by 10
    if <key [left arrow] pressed?> then
      change x by -10
    if <key [up arrow] pressed?> then
      change y by 10
    if <key [down arrow] pressed?> then
      change y by -10

-- Player sprite, fire script (replaces "when [space] key pressed"):
when green flag clicked
  set [can_shoot] to true
  forever
    if <<key [space] pressed?> and <can_shoot>> then
      create clone of [Bullet]
      start sound [sfx_shoot]
      set [can_shoot] to false
      wait 0.2 seconds
      set [can_shoot] to true
```